

# Oka's coherence theorem

**Source.** Freitag, *freitag.pdf*, PDF pp. 31–35 (printed pp. 28–32), Theorem 10.3 and Lemma 10.4.

## Statement

Let  $U \subseteq \mathbb{C}^n$  be open and let

$$F : \mathcal{O}(U)^p \longrightarrow \mathcal{O}(U)^q$$

be a holomorphic matrix. For every  $a_0 \in U$ , there are a neighborhood  $V \subseteq U$  of  $a_0$  and sections  $s^{(1)}, \dots, s^{(N)} \in \mathcal{O}(V)^p$  such that, for every  $a \in V$ ,

$$\ker F_a = \sum_{j=1}^N \mathcal{O}_a s_a^{(j)}. \quad (1)$$

Thus the sheaf of relations among finitely many holomorphic functions is locally finitely generated.

## 1. Reduction to one row

The case  $q = 0$  is immediate, and it suffices to prove the theorem when  $q = 1$ . Indeed, suppose it is known for target ranks  $q - 1$  and 1, and write

$$\alpha : \mathcal{O}^q \rightarrow \mathcal{O}^{q-1}, \quad \beta : \mathcal{O}^q \rightarrow \mathcal{O}$$

for the coordinate projections. Near a fixed point, choose  $A^{(1)}, \dots, A^{(m)}$  generating every stalk of  $\ker(\alpha F)$ , and let

$$G : \mathcal{O}^m \rightarrow \mathcal{O}^p, \quad G(h) = \sum_{r=1}^m h_r A^{(r)}.$$

After shrinking again, choose  $B^{(1)}, \dots, B^{(\ell)}$  generating the stalkwise kernel of the scalar row  $L = \beta FG$ . Then the sections  $G(B^{(s)})$  generate  $\ker F$ . In fact, they lie in  $\ker(\alpha F)$  and in  $\ker(\beta F)$ , hence in  $\ker F$ . Conversely, if  $x \in \ker F_a$ , then  $x = G_a(h)$  for some  $h$ , and  $0 = \beta F_a(x) = L_a(h)$ ; expressing  $h$  in the  $B_a^{(s)}$  expresses  $x$  in the  $G(B^{(s)})_a$ .

Induction on  $q$  now reduces everything to the scalar case.

## 2. Oka's bounded-degree lemma

Put

$$A = \mathbb{C}\{z_1, \dots, z_{n-1}\}, \quad B = A\{w\}, \quad P_{<m} = A[w]_{<m}.$$

Let  $F = (F_1, \dots, F_p) : B^p \rightarrow B$ , where every  $F_i \in P_{<d}$  and  $F_1$  is monic. Set

$$K = \ker F, \quad K_m = K \cap P_{<m}^p.$$

**Oka's lemma.** For every  $m \geq 3d$ ,

$$K = BK_m. \quad (2)$$

Here  $BK_m$  denotes the  $B$ -span of  $K_m$ .

### The Weierstrass case

First suppose that  $F_1$  is a Weierstrass polynomial. For  $j \geq 2$ , put

$$E^{(j)} = -F_j e_1 + F_1 e_j \in K_d.$$

Given  $G \in K$ , divide each coordinate by  $F_1$ :

$$G = F_1 A' + R, \quad A' \in B^p, \quad R \in P_{<d}^p.$$

Let  $S = \sum_i A'_i F_i$  and  $H = R + S e_1$ . A coordinate calculation gives

$$G = \sum_{j=2}^p A'_j E^{(j)} + H. \quad (3)$$

Moreover,

$$F(H) = 0, \quad F_1 S = - \sum_i F_i R_i \in P_{<2d}. \quad (4)$$

The second identity and uniqueness in Weierstrass division imply that  $S$  is a polynomial of degree  $< 2d$ . Hence  $H \in K_{2d}$ , while every  $E^{(j)} \in K_d \subseteq K_{2d}$ . Equation (3) proves

$$K = BK_{2d}. \quad (5)$$

### The monic case

For a merely monic  $F_1$ , Weierstrass preparation gives  $F_1 = QU$ , where  $Q$  is Weierstrass and  $U \in B^\times$ . Since  $F_1$  and  $Q$  are polynomials, uniqueness of division by  $Q$  shows that  $U \in A[w]$ ; moreover  $\deg Q, \deg U < d$ .

Let  $F' = (Q, F_2, \dots, F_p)$ ,  $K' = \ker F'$ , and

$$T(P_1, \dots, P_p) = (UP_1, P_2, \dots, P_p).$$

Then  $F'T = F$ , so  $T(K) = K'$ . By (5),  $K' = BK'_{2d}$ . Multiplication by the unit  $U$  does not change the span, so  $K'$  is also generated by the vectors  $UR, R \in K'_{2d}$ . Their inverse images are

$$T^{-1}(UR) = (R_1, UR_2, \dots, UR_p) \in K_{3d}.$$

Thus  $K = BK_{3d}$ , and  $K_{3d} \subseteq K_m \subseteq K$  proves (2) for every  $m \geq 3d$ .

## 3. Induction on the dimension: the scalar case

Assume the theorem in dimension  $n - 1$ , for all finite source and target ranks, and consider a row

$$F = (F_1, \dots, F_p) : \mathcal{O}(U)^p \rightarrow \mathcal{O}(U)$$

near  $a_0 \in U \subseteq \mathbb{C}^n$ . Translate  $a_0$  to the origin. Any  $F_i$  whose germ is zero vanishes after shrinking; split off its coordinate, with  $e_i$  as a kernel generator. We may therefore assume that all remaining germs are nonzero; if none remain, we are done.

Let  $h_i$  be the first nonzero homogeneous term of  $F_i$ . Choose  $v \in \mathbb{C}^n$  outside the finitely many proper algebraic sets  $h_i(v) = 0$ , and make  $v$  the last coordinate direction. The restriction of each  $F_i$  to the last coordinate line is then nonzero. Weierstrass preparation and a common shrinking give

$$F_i = Q_i U_i,$$

where  $Q_i$  is Weierstrass in  $z_n$  and  $U_i$  is a holomorphic unit. The diagonal automorphism

$$D(P_1, \dots, P_p) = (U_1 P_1, \dots, U_p P_p)$$

satisfies  $QD = F$  and identifies their kernel stalks. It is therefore enough to treat the row  $Q = (Q_1, \dots, Q_p)$ .

Shrink to a product  $V \times D_r \subseteq \mathbb{C}^{n-1} \times \mathbb{C}$ , and choose  $d$  larger than every  $z_n$ -degree of the  $Q_i$ . Fix  $m \geq 3d$ , set  $E_s = \mathcal{O}(V)[z_n]_{<s}$ , and consider

$$\Phi_m : E_m^p \longrightarrow E_{m+d}, \quad \Phi_m(P) = \sum_i Q_i P_i. \quad (6)$$

Taking coefficients identifies (6) with a holomorphic matrix

$$\mathcal{O}(V)^{mp} \longrightarrow \mathcal{O}(V)^{m+d}$$

on the  $(n-1)$ -dimensional base. By the induction hypothesis, after shrinking  $V$  its kernel is generated stalkwise by finitely many polynomial vectors

$$R^{(1)}, \dots, R^{(N)} \in E_m^p. \quad (7)$$

Fix  $a = (a', a_n) \in V \times D_r$  and put  $w = z_n - a_n$ . Translation from  $z_n$  to  $w$  preserves degree and monicity, and identifies the stalkwise kernel of (6) at  $a'$  with the bounded kernel  $K_m(a)$  over  $A = \mathcal{O}_{a'}$ . Thus the centered germs of (7) generate  $K_m(a)$  over  $\mathcal{O}_{a'}$ . Oka's lemma, applied in  $B = \mathcal{O}_{a'}\{w\} = \mathcal{O}_a$ , gives

$$\ker Q_a = \mathcal{O}_a K_m(a) = \sum_{s=1}^N \mathcal{O}_a R_a^{(s)}.$$

The same finite family works for every  $a \in V \times D_r$ . Transport it back through  $D^{-1}$  and the coordinate change, and restore the standard basis vectors belonging to the zero columns. This proves the theorem for one row in dimension  $n$ .

## 4. Completion of the proof

For  $n = 0$ , the assertion is ordinary finite-dimensional linear algebra. Assume it in dimension  $n - 1$  for every target rank. Section 3 proves the one-row case in dimension  $n$ , and Section 1 then proves every finite target rank. Induction on  $n$  proves (1).

## Notes on the source

- In the diagram on printed p. 30,  $\alpha$  must project to the first  $q - 1$  coordinates and  $\beta$  to the last coordinate. The scalar map is  $\beta FG$ , not  $\beta A$ .
- In the special Oka lemma, the conclusion is  $H \in K_{2d}$ .
- The  $3d$  argument must first replace  $R \in K'_{2d}$  by the equally generating vector  $UR$  before applying  $T^{-1}$ . One cannot generally infer that an inverse-unit multiple of a polynomial is polynomial.
- The last factor of the polydisc on printed p. 32 is a complex disc  $D_r$ , not the real interval  $(-r, r)$ .